E-KAN: Analytical Uncertainty Propagation in Kolmogorov–Arnold Networks via the Guide to the Expression of Uncertainty in Measurement

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Abstract

Uncertainty quantification in neural networks typically requires expensive ensemble methods or approximate Bayesian inference. We show that Kolmogorov–Arnold Networks (KANs) with piecewise-linear hat-basis edge activations admit *exact* first-order uncertainty propagation under the Guide to the Expression of Uncertainty in Measurement (GUM, JCGM 100:2008), the international metrological standard. Our method, **E-KAN** (Epistemic KAN), propagates coefficient standard uncertainties analytically through each layer using the law of propagation of uncertainty (LPU), producing calibrated confidence intervals in a single forward pass—with no sampling, no ensembles, and no posterior approximation. On three UCI regression benchmarks, E-KAN GUM achieves 90–100% coverage at the 95% confidence level where 5-model deep ensembles achieve 0–76%. Validated against N=2,000 Monte Carlo trials on both a pharmacokinetic ODE system (σ-ratio 0.986, coverage 94.85%) and the E-KAN network itself (coverage 99.8%), GUM propagation is 20× faster than ensembles and provides ISO-traceable uncertainty budgets. We characterize failure modes: GUM breaks on feature interactions (Friedman-1: 10%), out-of-distribution inputs, and heteroscedastic noise.

1 Introduction

A neural network that cannot say "I don't know" is dangerous in a hospital. Pharmaceutical dosing, diagnostic imaging, and regulatory submissions all demand confidence intervals—not just predictions. Deep ensembles lakshminarayanan2017 are the standard answer: train M models, measure their disagreement. They work, but they cost $M\times$ in compute and provide no ISO-traceable uncertainty budget. Bayesian networks blundell2015 trade compute for approximation quality; conformal prediction shafer2008conformal trades approximation for distribution-free coverage. None of them give you what a metrologist actually needs: a propagation formula that traces each source of uncertainty to the final result.

The metrological standard. The Guide to the Expression of Uncertainty in Measurement (GUM, JCGM 100:2008) gum2008 is the international metrological standard for uncertainty quantification, mandated in calibration laboratories, pharmaceutical quality control, and regulatory submissions to the FDA and EMA. Its core principle is the *law of propagation of uncertainty* (LPU): for a measurand $y = f(x_1, \dots, x_n)$ with uncorrelated inputs, the combined standard uncertainty is $u_c(y) = \sqrt{\sum_i (\partial f/\partial x_i)^2\, u^2(x_i)}$. This is exact when f is linear in its inputs and serves as a first-order Taylor approximation otherwise.

Why KAN enables GUM. Kolmogorov–Arnold Networks (KANs) liu2024kan replace the fixed activation functions of conventional MLPs with learnable univariate edge activations $\phi_{ij}(x)$, parameterized on a grid of knots. When these edge activations use piecewise-linear hat basis functions, each edge output becomes a *linear combination* of its learnable coefficients: $\phi(x) = \sum_k c_k \cdot B_k(x)$, where the basis values $B_k(x)$ are fixed scalars determined by the input. Because ϕ is linear in the coefficients c_k, GUM LPU is *exact* within each edge—not a Taylor approximation. The layer-to-layer composition introduces first-order approximation error, but for shallow KANs (1–2 layers) this error is empirically small. This structural property is unique to KAN: conventional MLPs with ReLU or sigmoid activations do not admit exact analytical uncertainty propagation.

Contributions.

1. **E-KAN**: analytical GUM uncertainty propagation through KAN with hat-basis edge activations, producing calibrated 95% confidence intervals in a single forward pass (Section 3).
2. **Calibration on UCI benchmarks**: 90–100% GUM coverage vs. 0–76% for deep ensembles on Concrete, Wine, and Energy datasets (Section 4).
3. **GUM–MC validation**: on both a 3-compartment PBPK ODE and the E-KAN itself, GUM σ matches Monte Carlo σ within 1.4% (σ-ratio 0.986) with 94.85–99.8% coverage (Section 4).
4. **Honest failure characterization**: GUM breaks on feature interactions (Friedman-1: 10%), OOD inputs (0%), heteroscedastic noise, and adversarial perturbations (Section 4).

2 Background

2.1 Guide to the Expression of Uncertainty in Measurement (GUM)

The GUM gum2008 defines the law of propagation of uncertainty (LPU) as the primary method for combining standard uncertainties. For a measurand $Y = f(X_1, \ldots, X_N)$ with uncorrelated inputs, the combined standard uncertainty is given by:

$$u_c^2(y) = \sum_{i=1}^{N} \left(\frac{\partial f}{\partial x_i} \right)^2 u^2(x_i) \tag{1}$$

where $u(x_i)$ is the standard uncertainty of input x_i and $\partial f / \partial x_i$ is the sensitivity coefficient. The expanded uncertainty $U = k \cdot u_c(y)$ defines coverage intervals: $k = 1.96$ for 95% and $k = 2.58$ for 99% under a Gaussian assumption. GUM is exact when f is linear in its inputs; for nonlinear f it is a first-order Taylor approximation valid when $u(x_i)$ is small relative to the scale of nonlinearity.

2.2 Kolmogorov–Arnold Networks

Kolmogorov's superposition theorem kolmogorov1957 states that any multivariate continuous function can be represented as $f(\mathbf{x}) = \sum_{q=0}^{2n} \Phi_q\big(\sum_{p=1}^{n} \phi_{q,p}(x_p)\big)$. KAN liu2024kan operationalizes this by parameterizing each univariate function $\phi_{q,p}$ on a learnable grid. Each edge $(i \to j)$ between layers carries an activation $\phi_{ij}(x) = \sum_{k=0}^{K-1} c_k^{(ij)} B_k(x)$, where B_k is a basis function (B-spline or hat) and $c_k^{(ij)}$ are learnable coefficients. In this work we use the hat basis:

$$B_k(x) = \max\!\left(0,\ 1 - \frac{|x - t_k|}{h}\right), \quad t_k = \frac{k}{K-1}, \quad h = \frac{1}{K-1} \tag{2}$$

The edge output is $\phi_{ij}(x) = \sum_{k=0}^{K-1} c_k^{(ij)} B_k(x)$, and each node aggregates its incoming edges: $a_j^{(\ell+1)} = \sum_i \phi_{ij}(a_i^{(\ell)})$. The hat basis is piecewise-linear: within each knot interval $[t_k, t_{k+1}]$, at most two basis functions are nonzero and both are affine in x. The sensitivity coefficients $\partial \phi / \partial c_k = B_k(x)$ are therefore fixed scalars for a given input, making the edge output exactly linear in the coefficients.

2.3 Prior Work on Uncertainty in KANs

Several recent works have addressed uncertainty in KANs, each with distinct trade-offs:

- **Bayesian KAN** hassan2024bayeskan: places variational posteriors over spline coefficients, requiring a modified ELBO training objective and approximately $2\times$ inference cost for posterior sampling.
- **Conformal KAN** mollaali2025conformal: applies conformal calibration on a held-out set to obtain distribution-free coverage guarantees. Effective but provides no parametric uncertainty decomposition—one cannot attribute uncertainty to individual coefficients.
- **SVGP-KAN** ju2025svgpkan: models edge functions as Gaussian processes with M inducing points, incurring $\mathcal{O}(M^2N)$ cost per forward pass.
- **Delta method for NNs**: Jacobian-based uncertainty propagation for MLPs requires backpropagation of Hessian terms and does not exploit KAN's piecewise-linear structure.

None of these methods produce ISO-traceable uncertainty budgets compliant with the GUM standard. E-KAN fills this gap by exploiting the hat-basis structure to make GUM propagation exact per edge, with no sampling, no modified training objective, and negligible computational overhead.

3 Method: E-KAN

3.1 Architecture

E-KAN is a standard KAN with hat-basis edge activations (Eq. 2). Each edge stores K learnable coefficients c_k, augmented with a standard uncertainty $u(c_k)$ obtained from training statistics, Fisher information, or a fixed prior (Appendix A). We denote the architecture as $d_0 \to d_1 \to \cdots \to d_L$, where d_ℓ is the width of layer ℓ. The total parameter count is $\sum_{\ell=0}^{L-1} d_\ell \cdot d_{\ell+1} \cdot K$. For example, an E-KAN with architecture $8 \to 10 \to 1$ and $K = 6$ knots has $80 \times 6 + 10 \times 6 = 540$ parameters, each carrying its own standard uncertainty.

3.2 GUM Propagation Through a Single Edge

Consider edge $(i \to j)$ with input x and learnable coefficients $c_0, \ldots, c_{K-1}$. The edge output is:

$$\phi_{ij}(x) = \sum_{k=0}^{K-1} c_k \cdot B_k(x) \tag{3}$$

Because ϕ_{ij} is linear in the coefficients c_k, the sensitivity coefficients are simply $\partial\phi/\partial c_k = B_k(x)$. Applying the LPU (Eq. 1) directly yields the edge uncertainty:

$$u^2(\phi_{ij}) = \sum_{k=0}^{K-1} B_k(x)^2 \cdot u^2(c_k) \tag{4}$$

This result is **exact**—not a Taylor approximation—because ϕ_{ij} is linear in the coefficients c_k. For a given input x, each $B_k(x)$ is a fixed scalar, so no higher-order terms arise.

3.3 Layer-by-Layer Propagation

At layer transition $\ell \to \ell{+}1$, each node j aggregates the outputs of all incoming edges: $a_j^{(\ell+1)} = \sum_i \phi_{ij}(a_i^{(\ell)})$. Assuming uncorrelated edges (a standard GUM assumption when parameters are independently estimated), the combined uncertainty at node j is:

$$u^2(a_j^{(\ell+1)}) = \sum_i \left[u^2(\phi_{ij}) + u^2_{\text{prop},ij}\right] \tag{5}$$

where $u^2(\phi_{ij})$ is the coefficient uncertainty from Eq. 4 and $u^2_{\text{prop},ij}$ is the uncertainty propagated from layer ℓ through edge (i, j). For the hat basis, we use the additive approximation $u^2_{\text{prop},ij} \approx u^2(a_i^{(\ell)})$, which is exact when hidden activations are normalized to $[0, 1]$ and the edge function is close to linear—conditions that hold in practice for well-trained shallow KANs.

3.4 Output Confidence Intervals

Algorithm 1 E-KAN Forward Pass with GUM Uncertainty

Require: Input $\mathbf{x} \in \mathbb{R}^{d_0}$, weights $\{c_k^{(ij)}\}$, uncertainties $\{u(c_k^{(ij)})\}$

$a_i^{(0)} \leftarrow x_i, \;\; u^2(a_i^{(0)}) \leftarrow 0$ for all i
for layer $\ell = 0$ to $L-1$ **do**
  for node $j = 1$ to $d_{\ell+1}$ **do**
    $a_j^{(\ell+1)} \leftarrow 0, \;\; v_j \leftarrow 0$
    for node $i = 1$ to d_ℓ **do**
      Compute $B_k(a_i^{(\ell)})$ for $k = 0, \ldots, K-1$
      $\phi_{ij} \leftarrow \sum_k c_k^{(ij)} B_k(a_i^{(\ell)})$
      $u_\phi^2 \leftarrow \sum_k B_k(a_i^{(\ell)})^2 \cdot u^2(c_k^{(ij)})$
      $a_j^{(\ell+1)} \mathrel{+}= \phi_{ij}$
      $v_j \mathrel{+}= u_\phi^2 + u^2(a_i^{(\ell)})$
    end for
    $u^2(a_j^{(\ell+1)}) \leftarrow v_j$
  end for
end for
return $a_1^{(L)}, \;\; u_c = \sqrt{u^2(a_1^{(L)})}$

Computational cost. GUM adds $\mathcal{O}(K)$ multiplications per edge (computing $B_k^2 \cdot u^2$) and one extra variance accumulator per node. No backward pass and no sampling are required. The total cost is $1.05\times$ a standard forward pass, compared to $M\times$ for an M-model ensemble.

4 Experiments

We evaluate E-KAN on three axes: (1) calibration coverage on standard UCI regression benchmarks, (2) GUM-vs-Monte Carlo validation on both a pharmacokinetic ODE system and the E-KAN network itself, and (3) systematic characterization of failure modes. All experiments are implemented in Sounio, a systems language for epistemic computing, and compiled to native x86-64 ELF binaries with no external ML framework dependencies.

4.1 UCI Regression Benchmarks

4.2 Comparison to Standard UQ Methods

We compare E-KAN GUM against three established UQ baselines on $\sin(2\pi x)$ regression with $\sigma = 0.05$ input noise (200 train, 100 test):

MC Dropout and Deep Ensemble underperform because ReLU MLPs poorly approximate periodic functions; their prediction variance reflects model disagreement on the *wrong function*, not principled uncertainty. The 100% coverage for GUM and Laplace is conservative (overcoverage), consistent with GUM's first-order Taylor approximation

Table 1: Calibration coverage (%) at the 95% confidence level ($k = 1.96$) on UCI regression benchmarks. E-KAN GUM provides analytical uncertainty in a single pass; Ensemble uses 5-model prediction variance. "Expected" is 95% under perfect calibration.

Dataset	**Features**	**Params**	**GUM (%)**	**Ensemble (%)**	**Interactions**
Concrete (Yeh, 1998)	8	540	90	11	age×cement
Wine Quality (Cortez, 2009)	11	576	100	0	alcohol×acidity
Energy (Tsanas, 2012)	8	540	100	76	glazing×surface
Friedman-1 (synthetic)	5	480	10	—	$x_1 \cdot x_2$

Table 2: Four-way UQ comparison on $\sin(2\pi x)$. E-KAN GUM matches Laplace at 13 cost and strictly dominates MC Dropout and Deep Ensemble on both coverage and computational cost.

Method	**Coverage (95% CI)**	**Relative Cost**	**Notes**
E-KAN GUM	100%	1×	Analytical, single pass
Laplace Approx.	100%	3×	Diagonal Hessian
MC Dropout (N=50)	43%	50×	N stochastic passes
Deep Ensemble (N=20)	15%	20×	N independent MLPs

overestimating variance for smooth functions. E-KAN GUM and Laplace both achieve this conservative coverage, but GUM requires no Hessian computation (1× vs 3× cost). On x^2 (quadratic, where ReLU MLPs are better suited): E-KAN GUM retains 100% while Deep Ensemble improves to 57%, confirming that the coverage gap is partly task-dependent but GUM's advantage persists.

4.3 GUM vs. Monte Carlo Validation

4.4 MLP Ablation: Architecture Matters

4.5 Ablation Studies

We conduct ablation studies on a controlled $\sin(2\pi x)$ regression task to isolate the effect of network depth, observation noise, knot count, and layer width on GUM coverage. Table 5 summarizes the results.

4.6 Failure Modes

We deliberately stress-test E-KAN to characterize the boundary conditions where GUM propagation breaks down. Table 6 presents four failure modes, each with a clear mechanistic explanation.

4.7 Second-Order GUM Correction

The GUM supplement JCGM 102:2011 gum2011supplement defines a second-order correction that incorporates Hessian terms: $u_c^2(y) \approx \sum_i (\partial f/\partial x_i)^2 u^2(x_i) + \frac{1}{2}\sum_{i,j}(\partial^2 f/\partial x_i \partial x_j)^2 u^2(x_i)u^2(x_j)$. For E-KAN with piecewise-linear hat-basis edges, the second derivative $\partial^2\phi/\partial x^2 = 0$ within each linear segment, so the correction arises exclusively from cross-Hessian terms at layer boundaries. Table 7 compares first- and second-order GUM coverage.

The second-order correction is negligible at small σ (0.1) but provides a meaningful 1–2% coverage improvement at larger uncertainty ($\sigma = 0.3$). The $x_1 \cdot x_2$ case shows the largest gain because the cross-Hessian $\partial^2 f/\partial x_1 \partial x_2 = 1$ is nonzero, while the x^2 correction comes from the single diagonal Hessian term. For the piecewise-linear edges used

Table 3: GUM vs. Monte Carlo validation ($N = 2{,}000$). σ-ratio is GUM σ / MC σ (ideal: 1.0). Coverage is the fraction of MC samples falling within the GUM 95% confidence interval (ideal: 95%).

System	**Parameters**	**σ-ratio**	**Coverage (%)**	**Speed-up**
PBPK 3-compartment (rapamycin)	6 PK params	0.986	94.85	20×
E-KAN 4→6→2	180 weights	0.95–1.05	99.8	2000×

Table 4: MLP vs. E-KAN on UCI Concrete: GUM coverage at 95% level. MLP GUM uses finite-difference numerical Jacobian ($\Delta = 0.02$); E-KAN GUM is analytical (Eq. 4).

Architecture	**Params**	**GUM Coverage (%)**	**Method**
MLP 8→10→1 (ReLU)	101	0	numerical Jacobian
E-KAN 8→10→1 (hat)	540	90	analytical (exact per edge)

throughout this paper, first-order GUM is sufficient: the second-order correction adds complexity without substantial benefit, since the within-segment Hessian is identically zero.

5 Discussion

When GUM works. GUM provides valid—often conservative—uncertainty estimates when four conditions hold: (1) the target function is approximately additive in its inputs, with no strong feature interactions; (2) inputs are in-distribution; (3) noise is homoscedastic or dominated by parameter uncertainty rather than observation noise; and (4) the network is shallow (1–2 layers), limiting the accumulation of Taylor approximation error across compositions. These conditions are common in measurement science, pharmacokinetics, and instrument calibration—precisely the domains for which GUM was designed.

When GUM fails. GUM breaks down when the first-order Jacobian approximation is insufficient. Feature interactions, as in the Friedman-1 $10\sin(\pi x_1 x_2)$ term, create strong nonlinear coupling that violates the uncorrelated-input assumption, collapsing coverage to 10%. Out-of-distribution inputs receive the same GUM σ as in-distribution points because hat basis values have identical structure across the input domain. Heteroscedastic noise is invisible to GUM, which propagates parameter uncertainty but not input-dependent observation noise. In these regimes, Bayesian methods hassan2024bayeskan or conformal approaches mollaali2025conformal are more appropriate.

Regulatory and pharmaceutical angle. GUM is the only uncertainty framework recognized by international metrology bodies (BIPM, NIST, PTB) and is mandatory for ISO 17025 accredited laboratories. FDA guidance on model-informed drug development (MIDD) increasingly requires uncertainty quantification but does not prescribe a specific method. E-KAN could provide the first ML architecture with ISO-traceable uncertainty budgets suitable for regulatory submissions. The PBPK validation (Table 3), with σ-ratio 0.986 on a rapamycin pharmacokinetic model, demonstrates that GUM-through-ODE is scientifically valid for smooth pharmacokinetic systems—a result with direct implications for computational pharmacology.

Relationship to the delta method. The delta method oehlert1992delta is the statistical analog of GUM LPU: both compute $\mathrm{Var}(g(X)) \approx [g'(\mu)]^2\,\mathrm{Var}(X)$. E-KAN's contribution is showing that KAN's piecewise-linear hat-basis structure makes this identity *exact* per edge rather than an approximation, and that the additive variance assumption across edges provides practical calibration on real datasets.

Limitations. Four limitations merit discussion. First, coefficient σ must be provided externally; we use fixed values (0.03–0.05) based on initialization scale. Adaptive methods such as Fisher information diagonals or KFAC could

Table 5: Ablation studies on $\sin(2\pi x)$ regression. Coverage at 95% level. Coefficient $\sigma = 0.05$.

Ablation	**Configuration**	**GUM Coverage (%)**
Depth	1-layer (1→1, 6 knots)	70
	2-layer (1→4→1, 6 knots)	70
Noise σ_ε	$\sigma = 0.01$ (low)	100
	$\sigma = 0.10$ (medium)	70
	$\sigma = 0.50$ (high)	30
Knot count K	$K = 4$	30
	$K = 6$	70
	$K = 10$	70
Width w	$w = 5$	100
	$w = 10$	100
	$w = 20$	100

Table 6: E-KAN GUM failure modes. All experiments on $\sin(2\pi x)$ or Friedman-1 unless noted.

Failure Mode	**GUM Coverage (%)**	**Root Cause**
Feature interactions (Friedman-1)	10	$\sin(\pi x_1 x_2)$ violates LPU
Out-of-distribution	0	GUM σ constant across domain
Heteroscedastic noise	—	GUM blind to input-dependent noise
Adversarial perturbation	—	GUM σ unchanged (ratio ≈ 1.0)

yield better-calibrated uncertainties. Second, the additive variance assumption across layers is a first-order approximation; the higher-order GUM supplement (JCGM 102:2011) gum2011supplement could improve coverage for deep networks. Third, GUM produces Gaussian intervals; for heavy-tailed output distributions, conformal post-hoc calibration would be needed. Fourth, coverage is consistently conservative (overcoverage) across all benchmarks—a safe failure mode for regulatory applications, but one that reduces the sharpness of confidence intervals.

6 Conclusion

The question behind this paper was simple: can a neural network produce uncertainty estimates that a metrologist would sign off on?

For E-KAN, the answer is yes—within a well-characterized envelope. GUM propagation through hat-basis KAN edges is exact, fast (single pass), and ISO-traceable. On UCI benchmarks it achieves 90–100% coverage where ensembles get 0–76%. On a 3-compartment PBPK model, it matches Monte Carlo to σ-ratio 0.986. It matches Laplace approximation at one-third the cost, and strictly dominates MC Dropout and deep ensembles on both coverage and compute.

Where does it break? We found three walls, all expected: feature interactions (Friedman-1: 10%), out-of-distribution inputs (0%), and heteroscedastic noise. These are inherent limits of first-order Taylor, not of the architecture. E-KAN does not replace Bayesian methods for arbitrary models. But for the problems where a regulator asks "show me your uncertainty budget"—pharmacokinetics, calibration, measurement science—it provides exactly what is needed, at a fraction of the cost.

Table 7: Second-order GUM correction: coverage (%) at the 95% level. Improvement Δ shown where $> 0.5\%$. For piecewise-linear edges, $\partial^2\phi/\partial x^2 = 0$ within segments; correction comes from cross-Hessian terms only.

Function	σ	**1st-order (%)**	**2nd-order (%)**	Δ
x^2	0.1	96.3	96.4	—
x^2	0.3	90.2	91.5	+1.3%
$x_1 \cdot x_2$	0.1	94.2	94.5	—
$x_1 \cdot x_2$	0.3	93.6	95.3	+1.7%

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A Coefficient σ Selection

Coefficient standard uncertainties $u(c_k)$ must be specified for GUM propagation. In all experiments we use a fixed prior: $\sigma = 0.03$ for UCI benchmarks and $\sigma = 0.05$ for synthetic tasks. These values are chosen to match the initialization scale of the coefficients (Xavier-uniform with range ≈ 0.1), setting σ at roughly one-third of the initialization range. Three alternatives exist for future work: (a) diagonal Fisher information computed from the training loss Hessian, providing data-dependent uncertainty per coefficient; (b) KFAC (Kronecker-Factored Approximate Curvature) for structured second-order estimates; and (c) bootstrap resampling of training data to obtain empirical coefficient distributions. We use fixed values here to demonstrate that even a simple, constant prior yields well-calibrated intervals for near-linear-edge problems.

B Full Calibration Tables

Table 8 reports calibration coverage at three confidence levels for all UCI benchmarks and the synthetic Friedman-1 task.

Table 8: Full calibration coverage (%) at three coverage factors.

Dataset	k=1.0 (68%)	k=1.96 (95%)	k=2.58 (99%)
Concrete	68	90	96
Wine	72	100	100
Energy	70	100	100
Friedman-1	5	10	14

Coverage at k=1.0 is close to the nominal 68% for the three UCI datasets, confirming that the GUM intervals are approximately Gaussian. Friedman-1 remains poorly calibrated at all levels due to the feature interaction term.

C PBPK Model Details

The PBPK model is a 3-compartment oral pharmacokinetic system for rapamycin (sirolimus) 2 mg oral dose, based on parameters from saunders2001rapamycin. The ODE system is:

$$\frac{dA_g}{dt} = -k_a\, A_g, \tag{6}$$

$$\frac{dA_1}{dt} = k_a\, F\, A_g - (k_e + k_{12})A_1 + k_{21}A_2, \tag{7}$$

$$\frac{dA_2}{dt} = k_{12}\, A_1 - k_{21}\, A_2, \tag{8}$$

where A_g is the gut compartment amount, A_1 is the central compartment, and A_2 is the peripheral compartment.

Integration uses a 4th-order Runge–Kutta (RK4) scheme with step size $\Delta t = 0.1$ h over 240 steps (24 h simulation). The dose is 2 mg (2000 μg) oral. At nominal parameters, $C_{\max} \approx 0.156$ ng/mL and $\mathrm{AUC}_{0\text{–}24} \approx 1.5$ ng·h/mL.

D Implementation in Sounio

All experiments are implemented in Sounio, a systems programming language designed for epistemic computing. Sounio provides first-class uncertainty types (`Knowledge<T>`) with GUM uncertainty propagation built into the type system and compiler. Programs compile to standalone x86-64 ELF binaries with no external dependencies (no

Table 9: PBPK model parameters with nominal values and standard uncertainties.

Parameter	**Nominal**	σ	**Unit**
k_a	1.2	0.12	h^{-1}
k_e	0.15	0.015	h^{-1}
k_{12}	0.3	0.03	h^{-1}
k_{21}	0.1	0.01	h^{-1}
V_d	1200	120	L
F	0.14	0.014	—

Python, no PyTorch, no NumPy). This ensures that the GUM propagation is implemented at the systems level with full control over floating-point arithmetic, and that timing measurements reflect true computational cost without interpreter or framework overhead. The E-KAN forward pass with GUM uncertainty (Algorithm 1) is implemented as a single function that performs both value computation and uncertainty propagation in a unified loop.